

Python Interview Guide

2 Marks, 5 Marks & Important Programs

PART A – 2 Marks Questions

- 1. What is Python?** Python is a high-level, interpreted, object-oriented programming language used for simple and powerful application development.
- 2. What are Python features?** Simple syntax, portability, dynamic typing, large library support, and open-source.
- 3. What is PEP 8?** It is the official style guide for writing clean and readable Python code.
- 4. What is a list?** A mutable collection that stores multiple values in a single variable.
- 5. What is a tuple?** An immutable collection used to store multiple values.
- 6. What is dictionary?** A collection that stores data in key–value pairs.
- 7. What is indentation?** Spaces used to define code blocks in Python.
- 8. What is a function?** A reusable block of code that performs a specific task.
- 9. What is an exception?** An error that occurs during program execution.
- 10. What is pip?** A package manager used to install Python libraries.

PART B – 5 Marks Questions

1. Explain Python and its features.

Python is a high-level programming language known for its simplicity and readability. It supports object-oriented, functional, and procedural programming. Features include platform independence, automatic memory management, large standard libraries, and strong community support.

2. Explain data types in Python.

Python supports several data types such as int, float, string, list, tuple, set, and dictionary. Each data type is used to store different kinds of values and helps in efficient memory usage.

3. Explain functions in Python.

Functions are blocks of reusable code defined using the def keyword. They reduce redundancy and improve modularity. Python supports built-in and user-defined functions.

4. Explain exception handling.

Python uses try, except, else, and finally blocks to handle runtime errors. This prevents program crashes and improves reliability.

5. Explain OOP concepts in Python.

Python supports object-oriented programming using classes and objects. The main concepts are Encapsulation, Inheritance, Polymorphism, and Abstraction.

PART C – Important Python Programs

1. Reverse a String

```
s = "python"
print(s[::-1])
```

2. Check Palindrome

```
s = "madam"
print(s == s[::-1])
```

3. Fibonacci Series

```
a, b = 0, 1
for i in range(10):
    print(a, end=" ")
    a, b = b, a+b
```

4. Factorial

```
n = 5
fact = 1
for i in range(1, n+1):
    fact *= i
print(fact)
```

5. Prime Number

```
n = 7
flag = True
for i in range(2, n):
    if n % i == 0:
        flag = False
print(flag)
```

6. Largest of Three Numbers

```
a,b,c = 10,20,15
print(max(a,b,c))
```

7. Count Vowels

```
s="hello"
vowels="aeiou"
count=0
for ch in s:
    if ch in vowels:
        count+=1
print(count)
```

8. Reverse a Number

```
n=123
rev=0
while n>0:
    rev = rev*10 + n%10
    n//=10
print(rev)
```

9. Armstrong Number

```
n=153
temp=n
s=0
while temp>0:
    d=temp%10
    s+=d*d*d
    temp//=10
print(s==n)
```

10. Word Frequency

```
s="python is python"
d={}
for w in s.split():
```

```
    d[w]=d.get(w,0)+1  
print(d)
```